Reducing Observational Astronomy Time Costs with Multi-Slit Spectrographs

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Improving Efficiency with Multi-Slit Observations

Why multi-slit?

The Milky Way Bulge extra-tidal star survey **(MWBest)** aims to explore the origins of old stars near the Milky Way's center, hypothesizing that these stars were stripped from globular clusters. These clusters are groups of gravitationally bound stars moving at similar velocities.

Our analysis requires observing many stars to identify potential cluster members. To reduce costs and increase efficiency, we use a multi-slit spectrograph to observe multiple stars simultaneously **(Figures 1 and 2).**

What is multi-slit observation?

It involves using several slits on the same frame that let starlight pass through for data collection **(Figure 1).**

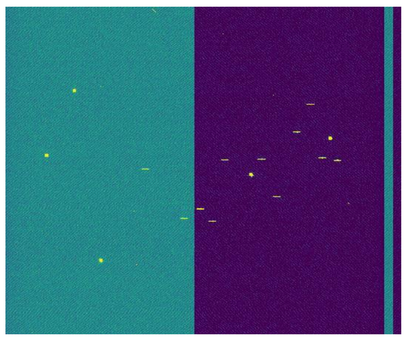

Figure 1: The horizontal lines are the slits, and the square boxes are reference stars for positioning the slits.

Field of stars

The telescope's slits target a field of stars **(Figure 2),** with potential cluster members.

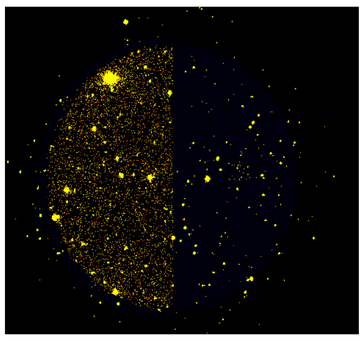

Figure 2: This is how a field of stars would appear when viewed without the spectrograph.

Data Collection

Our observations are conducted with the 3.5m Astrophysical Research Consortium **(ARC)** telescope **(Figure 3)** at Apache Point Observatory **(APO)** in New Mexico.

We use the Kitt Peak Ohio State Multi-Object Spectrograph **(KOSMOS)**, originally installed on the Mayall 4m telescope at Kitt Peak in 2013 and re-installed at APO in 2021.

Since KOSMOS lacks public data reduction software, **I developed a pipeline and guide for multi-object spectroscopy data reduction for KOSMOS. The software is called astro-pyvista and was built by Jon Holtzman.**

Figure 3: ARC 3.5m telescope at APO. Image by Anna Panova.

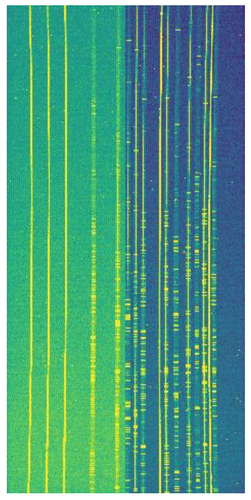

Figure 4: A raw science image from KOSMOS.

Building a Software Pipeline for KOSMOS Multi-Slit Reduction

The data reduction pipeline corrects factors affecting data quality.

Calibration steps include:

- **Bias subtraction:** adjusts for camera chip offset voltage
- **Dark subtraction (Figure 5):** removes thermal noise
- **Flat fielding (Figure 6):** corrects for nonuniformity of light
- **Cosmic ray removal:** removes high-speed particles registered as light
- **Sky background removal:** subtracts atmospheric emission lines

The pipeline traces and extracts each star's spectrum, calibrates wavelength, and allows adjustments, if needed.

Wavelength calibration does pixel-to-wavelength conversions using known emission lines from **calibration (arc) lamps (Figures 8 and 9).**

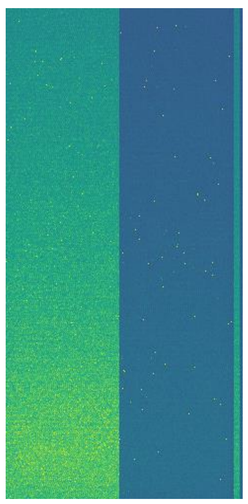

Figure 5: Dark image for thermal noise reduction.

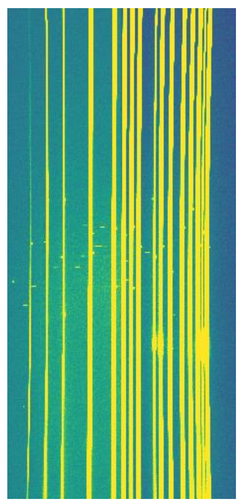

Figure 6: Flat image for flat fielding.

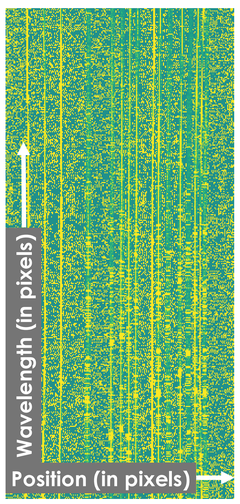

Figure 7: Reduced science image.

Interactive wavelength calibration

Assign each pixel a specific wavelength, then sample across the spectrum for a complete wavelength solution to calibrate a star in the same slit position as the arc spectrum.

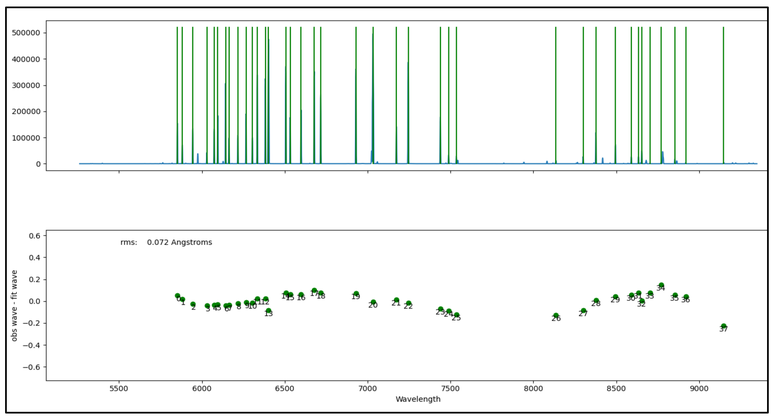

Figure 8: Calibrated spectrum of a slit of an arc lamp **(Figure 9).**

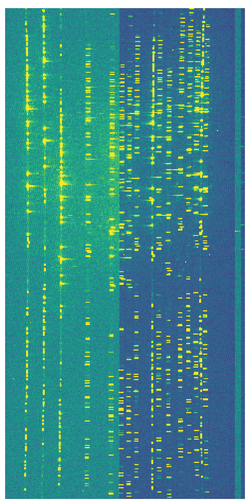

Figure 9: Arc lamp image (neon).

Results, Validation, and Future Prospects

The pipeline yields stellar spectra for data analysis **(Figure 10)**. To verify wavelength calibration, I reduced two APOGEE catalog stars with known radial velocities **(RV) (Table 1).**

Stellar spectra: each colored spectrum represents a specific star in a specific slit. The top plot shows spectra with the sky background subtracted, while the bottom plot displays the subtracted sky background.

The APOGEE catalog is the Apache Point Observatory Galactic Evolution Experiment, covering a unique footprint of 14,055 square degrees of sky.

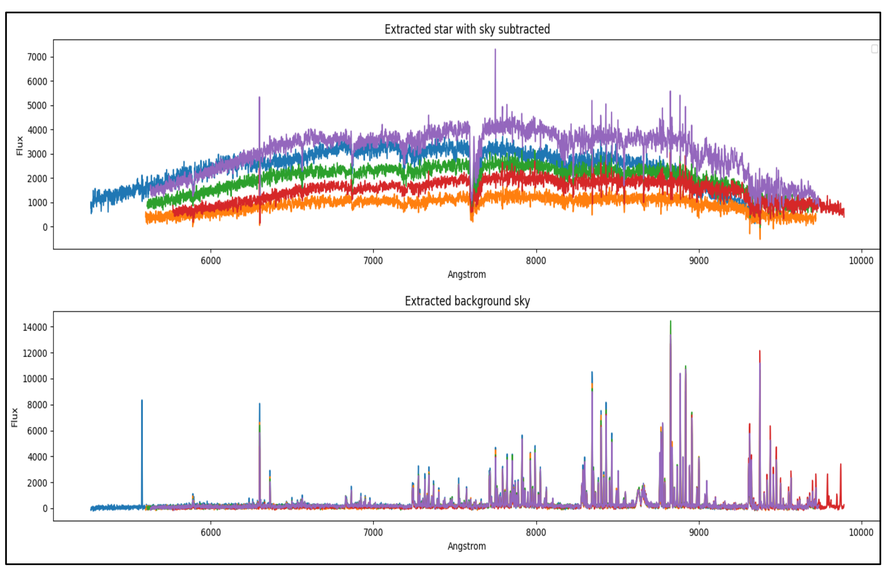

Figure 10: Slits extracted spectra.

Reduction Validation via APOGEE Stars with Known RVs

ID	Our RV (km/s)	VHELIO (km/s)	STARFLAG
2M16483173-0242328	-48.8	-38.2	0: bad pixels (>20%)
2M16483432-0240520	51.4	22.7	4: low S/N (S/N<5)

Table 1: Details of the two APOGEE stars. VHELIO (measured APOGEE RV), STARFLAG (APOGEE Flag details).

Our analysis:

KOSMOS has a velocity limit uncertainty of **32 km/s** per pixel.

✓ The first star's RV is **-48.8 km/s** with an approximate accuracy of **-10.6 km/s** and with **bad pixels (<5%)**.

? The second star's RV is **51.4 km/s** with a deviation of **28.7 km/s**. Although our **signal-to-noise ratio (S/N) of 37.487** suggests better data accuracy, further observations are required for a definite answer.

Note: Scan the **QR code** for the software guidelines, which are currently under review.

Future goals

- Reduce stars with known radial velocities and metallicities to validate our calibration further.
- Upload the reduction guidelines to APO Wiki for future users of multi-slit spectroscopy.

Acknowledgments

This work was funded through support from the M.J. Murdock Charitable Trust's RAISE program. Our RAISE Milky Way Bulge extra-tidal star survey (MWBest) collaborators are Dr. Joanne Hughes et al. (Seattle University), Dr. Andrea Kunder et al. (St. Martin's University), and Dr. Kevin Covey et al. (Western Washington University).

References and citations

Pace et al. 2023, MNRAS, 526, 1075
✷ Rich et al. 2020, MNRAS, 499, 2340.✷ pyvista by Jon Holtzman (Welcome to pyvista's documentation! — pyvista v0.4.0 documentation). ✷APOGEE catalog (Imaging Data Overview | SDSS).